Integrating the Triadic Synthesis Framework into the GRAPHIA SSH Knowledge Graph

A Federated Governance Layer for Community-Validated Linguistic Data

Technical Whitepaper — Proposal for WP2/WP4 Integration

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1. Executive Summary

The GRAPHIA Data Acquisition Platform (DAP) currently commits entities and relations extracted from full-text documents by WP4 AI/NLP services directly into the GoTriple Knowledge Graph, with no step for human review, cultural verification, or contributor attribution. This is adequate for high-resource, well-represented content, but it systematically underserves low-resource languages, minority dialects, and culturally specific material — precisely the long tail that the SSH Knowledge Graph is meant to represent.

This document proposes integrating the Triadic Synthesis Framework (Human Contributors + Agentic Scaffolding + DAO-based Governance, described in the companion academic paper) as a federated validation layer for this content, connected to GRAPHIA through existing extension points rather than through changes to the authoritative RDF core.

The proposal does not modify Virtuoso, the TRIPLE Ontology, or any accepted architectural decision (ADR002). It adds one new DAP microservice, one ontology extension namespace, and one external federation node, all reversible and independently governed.

2. Motivation: The Validation Gap in WP4 Entity Extraction

Section 4.4.5 of the GRAPHIA Technical Architecture (D2.2) describes entity and relation extraction as a fully automated WP4 service: entities are linked to triple:Document instances via schema:mentions and an oa:Annotation body, with oa:creator identifying the extracting service. No mechanism exists for a domain expert or an affected community to confirm, correct, or contest an extracted fact before it is published.

This mirrors what knowledge engineering has long called the input bottleneck (Feigenbaum, 1977): high-quality structured data depends on a narrow pool of expert annotators. GRAPHIA has effectively removed the bottleneck by removing the experts — which trades scale for reliability, particularly for:

Minority and endangered languages, where generic NLP models trained on high-resource corpora perform poorly;

Culturally specific entities (local place names, informal titles, oral-tradition references) that general-purpose entity recognisers systematically mis-tag or miss;

Content where provenance and contributor attribution carry ethical weight — the concern this framework refers to as data alienation.

The Triadic Synthesis Framework was designed to address exactly this gap: it restores a human validation step without reintroducing the centralised editorial bottleneck that made expert-only curation unscalable in the first place.

3. Proposed Integration: A Federated Node, Not a Core Change

GRAPHIA's own architecture is built around federation rather than centralised ingestion (D2.2, Section 2.1): the SSH KG connects autonomous graphs (OpenCitations, EHRI, GESIS, ORKG) through the GRAPHIA Ontology and the SKG-IF interoperability layer, each graph retaining its own governance, schema evolution, and operational rules.

This proposal treats the Federated Epistemic Node (FEN) — the system implementing the Triadic Synthesis Framework — as exactly this kind of participant: an autonomous node with its own DAO governance, reputation system, and validation infrastructure, connected to GRAPHIA only through message-passing and dereferenceable identifiers. No component of FEN's governance stack (DAO, Quadratic Voting, reputation ledger) needs to be hosted, operated, or maintained by GRAPHIA partners.

Three integration surfaces are used, all of which already exist in the D2.2 architecture:

The Data Acquisition Platform's event bus (Kafka) — one additional consumer/producer pair, following the same asynchronous, non-blocking pattern already used for translation, classification, and annotation services (Section 4.2).

The TRIPLE Ontology's provenance mechanism — named graphs plus oa:/PROV-style properties (Section 3.4–3.5) — extended with a small, additive namespace rather than modified.

The Persistent Identifier scheme (Section 4.5) — extended with a dedicated namespace for governance artefacts, following the same ARK + w3id.org pattern already adopted for GoTriple KG resources.

4. Technical Architecture

4.1 Ontology Extension: the gfen: Namespace

A single additive namespace records validation state and governance provenance on existing oa:Annotation instances, without altering triple:Document, triple:Profile, or any core class.

```
@prefix gfen: <https://w3id.org/got/fen/ontology#> .

gfen:ValidationStatus a rdfs:Class .
gfen:pending    a gfen:ValidationStatus .
gfen:validated  a gfen:ValidationStatus .
gfen:disputed   a gfen:ValidationStatus .
gfen:rejected   a gfen:ValidationStatus .

gfen:validationStatus      a rdf:Property .  # -> gfen:ValidationStatus
gfen:validationMethod      a rdf:Property .  # -> gfen:QuadraticVoting | gfen:PeerReview
gfen:governanceDecisionId  a rdf:Property .  # -> dereferenceable PID (see 4.3)
gfen:reputationSnapshot    a rdf:Property .  # -> hash of vote-weight state
gfen:ledgerAnchor          a rdf:Property .  # on-chain tx hash, anchor only
gfen:contributorProfile    a rdf:Property .  # -> triple:Profile
```

Contributor identity is not re-modelled: FEN reuses the existing triple:Profile class, avoiding a duplicate identity layer between GRAPHIA and FEN.

4.2 Event-Driven Validation Flow

A single new DAP microservice — the FEN Bridge — is inserted between Entity & Relation Extraction and the Publisher. It does not block publication: candidate entities are published immediately with status gfen:pending, exactly as they are today. Validation outcomes arrive asynchronously and update the same named graph in place.

```
doc.normalized.v1
     |
     v
[ Entity & Relation Extraction ]   (existing WP4 service)
     | emits candidates, status = pending
     v
dap.entities.pending_validation.v1
```

```
     |
     v
[ FEN Bridge ]                     (new DAP microservice)
     | forwards batches to external FEN API
     v
  [ FEN: Agentic Scaffolding (ElizaOS) -> DAO / Quadratic Voting ]
     | governance decision, asynchronous callback
     v
fen.governance.decisions.v1
     |
     v
[ Validation Result Consumer ]     (new DAP microservice)
     | str_replace of gfen:validationStatus in the named graph
     v
dap.entities.validated.v1
     |
     v
[ Publisher ]                       (existing service, unchanged)
     |
     v
Virtuoso (GoTriple KG)  +  GoTriple indexes
```

This preserves the eventual-consistency principle stated in Section 4.1 of D2.2: if a candidate entity is never voted on, it remains published with status gfen:pending indefinitely — there is no failure mode in which the pipeline stalls.

4.3 Persistent Identifiers for Governance Records

Governance decisions receive their own dereferenceable identifiers, following the ARK + w3id.org pattern already defined for the GoTriple KG (D2.2, Section 4.5), under a dedicated Name Assigning Authority Number (NAAN) rather than the GoTriple namespace.

```
ark:{FEN_NAAN}/g00042        # governance decision
ark:{FEN_NAAN}/v00042        # validation record (entity + decision + snapshot)
ark:{FEN_NAAN}/r00042        # reputation snapshot at decision time

https://n2t.net/ark:{FEN_NAAN}/g00042
   -> redirects to ->
https://w3id.org/fen/id/decision/g00042
```

Only an aggregated decision record is published at this PID — quorum, outcome, method, and a link to the reputation snapshot — not individual votes. This keeps the governance record GDPR-compatible: no personal voting data is exposed, and only the on-chain anchor (a hash) is immutable, not the underlying content in Virtuoso.

5. Architecture Decisions

Two decisions define the boundary of this proposal and directly address the objections a reviewer would expect to raise first.

6. Risk Assessment and Mitigations

7. Request to the Consortium

To move from this design to a working MVP, the following are requested from GRAPHIA partners:

A dedicated NAAN (or a sub-range under an existing one) for FEN-issued PIDs, following Section 4.5's registration process.

Read access to a test-environment Kafka topic mirroring dap.entities.pending_validation.v1, scoped to a single low-resource-language WP4 test corpus.

Confirmation that the gfen: namespace extension (Section 4.1) does not require GRAPHIA Ontology governance sign-off, since it is additive and does not modify triple:* classes.

A point of contact within WP4 to align on the exact schema of extracted-entity messages, so the FEN Bridge consumes them without transformation.

8. Next Steps

MVP: FEN Bridge microservice + single test corpus + manual DAO quorum, demonstrated end-to-end against a GRAPHIA test instance.

Evaluation: compare entity precision/recall before and after community validation on the test corpus.

Federation: formal registration of FEN as an external node in the SSH KG federation (Section 2.1), alongside OpenCitations, EHRI, GESIS, and ORKG.

Publication: joint reporting of results as part of WP2/WP4 deliverables, subject to consortium agreement.

Companion document: "Decentralised Agentic Governance: A Methodology for Community-Owned Linguistic Datasets and Knowledge Synthesis" (Riznyk, 2026) provides the full theoretical framework underlying this proposal.

	Rationale
content remains in Virtuoso.	Avoids conflict with ADR002 (Virtuoso as authoritative RDF core) and preserves GDPR right-to-erasur
n), not embedded inside GoTriple KG or the DAP core.	Matches GRAPHIA's own federation philosophy; requires no GRAPHIA partner to operate or govern D

	Mitigation
multiple identities to amplify voting power.	Contributor identity reuses triple:Profile; identity verification strength is a FEN-side governance paramete
umulate disproportionate reputation-based influence.	Reputation decay and Quadratic Voting cost structure (n^2) cap the influence of any single actor; paramet
not be deleted once written.	Only decision hashes are anchored on-chain (ADR-FEN-001); all deletable content stays in Virtuoso unc
y validation could block publication.	Candidates publish immediately as gfen:pending; validation updates arrive asynchronously (Section 4.2)